

Build a Data Mart in SQL

Phase 3 - Finalization Phase

Abstract

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Abstract

The Airbnb Datamart project was focused on validating and improving the database to efficiently manage key functions of an Airbnb-like platform. Created with MySQL Workbench, the database is designed to ensure strong performance, accuracy, and data integrity while supporting the essential operations of the platform.

Database Structure and Design

The database was built using an Entity Relationship Model (ERM) with 22 entities representing the core parts of the platform, such as Guests, Hosts, Bookings, Payments, Reviews, and Vacation Rentals. Relationships between these entities are managed through foreign keys, while recursive relationships allow hosts to refer to other hosts. Triple relationships between Guest, Booking, and Transaction ensure the payment process is accurately captured. Normalization was applied to make data storage more efficient and reduce redundancy. For example, the City entity manages location data and links to the Location and VacationRental tables. Refinements in the database included improving the representation of payments and refunds and allowing both guests and hosts to leave reviews, creating a full record of user interactions.

Core Database Functionalities

The database supports the main roles of the platform, enabling guests, hosts, and admins to perform their tasks smoothly. The key functionalities include:

- **User Management:** Guests and hosts can manage their profiles, personal information, and preferences, while TravelAdmins handle account management and customer support.
- **Property Listings:** Hosts can list properties with all the necessary details. The Location table stores address information to ensure accurate location data for each rental.
- **Booking Management:** Guests can book properties, and their reservations are recorded in the Booking table, which tracks dates, payment status, and price. Hosts can check availability and view booking history.
- **Payment and Transaction Processing:** Payments are managed via the Transaction table, linking each payment to a booking, tracking payment status, and ensuring financial transactions are handled securely.
- **Review System:** Both guests and hosts can submit reviews through the Review table, with each review connected to the relevant booking, capturing feedback from both sides.

- Customer Service Management: The CustomerService table tracks issues, resolutions, and contact methods, allowing admins to manage customer service requests effectively.

Data Population and Testing

Each of the 22 tables was populated with at least 20 records to reflect real-world scenarios, such as booking rentals, completing transactions, submitting reviews, and handling customer service issues. This test dataset enabled thorough validation of the database's functionality.

Test cases verified that relationships between entities worked as expected, such as guests booking properties, completing payments, and leaving reviews. Additional tests confirmed that booking cancellations and refunds were handled properly, simulating real payment flows. Cancellations and refunds were also tested to make sure the payment system worked properly.

Indexes on important fields like BookingID and TransactionID optimized query performance, ensuring the database can grow and handle larger datasets effectively.

Design Enhancements and Scalability

Several important improvements were made throughout the project. The payment system was updated to better reflect real-life transactions, and both guests and hosts are now able to leave reviews. The Location table was expanded to include detailed address information (street, city, state, and country), to reduce redundancy and improve efficiency while allowing for more precise location management and easier property searches by linking it to other tables via foreign keys. The database was designed to scale with the platform, meaning it can handle more users, bookings, and properties without losing performance or data accuracy.

Conclusion

The Airbnb Datamart project successfully developed a strong and scalable database system that can handle key platform operations, such as managing users, properties, bookings, payments, and reviews. Relationships between entities were carefully structured to ensure data integrity and seamless performance. Test cases confirmed that the system can handle real-world scenarios, showing that the database functions as expected. It is now ready for future improvements or deployment, providing a solid base for an Airbnb-like platform.

1. Metadata Overview

In this section, key metadata insights about the schema are presented. The analysis includes information on the structure of the tables, the size of the schema, and the foreign key constraints, all essential for ensuring that the database operates efficiently and securely. This section provides an in-depth look into how the database is structured and how it maintains data integrity.

Column Metadata for Each Table:

The schema consists of 22 tables, each containing at least 20 entries. This metadata shows a clear system structure, with each table logically connected.

TABLE NAME	COLUMN NAME	DATA TYPE	IS NULLABLE	COLUMN KEY	EXTRA
amenity	AmenityID	int	NO	PRI	auto_increment
amenity	AmenityName	varchar	YES		
amenity	Description	text	YES		
booking	BookingID	int	NO	PRI	auto_increment
booking	GuestID	int	NO	MUL	
booking	RoomID	int	NO	MUL	
booking	BookingDate	datetime	NO		
booking	CheckInDate	date	NO		
booking	CheckOutDate	date	NO		
booking	TotalPrice	float	NO		
booking	PaymentStatus	enum	NO		
booking	LengthOfStay	int	NO		
booking	CancellationDeadline	date	YES		
booking	CancellationRefund	float	YES		
booking	DateOfCancellation	date	YES		
booking	HostPayout	float	YES		
booking	EventName	varchar	YES		
booking	StartDate	date	YES		
booking	EndDate	date	YES		
booking	Description	text	YES		
cancellationpolicy	PolicyID	int	NO	PRI	auto_increment
cancellationpolicy	PolicyName	varchar	YES		
cancellationpolicy	Description	text	YES		
city	CityID	int	NO	PRI	auto_increment
city	CityName	varchar	YES		
city	Country	varchar	YES		
customerservice	CustomerServiceID	int	NO	PRI	auto_increment
customerservice	BookingID	int	NO	MUL	
customerservice	IssueDescription	text	YES		
customerservice	Resolution	text	YES		
customerservice	ContactMethod	varchar	YES		
customerservice	ResolutionDate	date	YES		
event	EventID	int	NO	PRI	auto_increment
event	BookingID	int	NO	MUL	
event	EventName	varchar	YES		
event	StartDate	date	YES		
event	EndDate	date	YES		
event	Description	text	YES		
guest	GuestID	int	NO	PRI	auto_increment

guest	Name	varchar	NO		
guest	Email	varchar	NO	UNI	
guest	Password	varchar	NO		
guest	PhoneNumber	varchar	YES		
guest	ProfilePicture	varchar	YES		
guest	Street	varchar	YES		
guest	City	varchar	YES		
guest	State	varchar	YES		
guest	Country	varchar	YES		
guest	GDPRAcknowledgement	tinyint	YES		
guest	LanguageSettings	varchar	YES		
guestsocialnetwork	GuestSocialNetworkID	int	NO	PRI	auto_increment
guestsocialnetwork	GuestID	int	NO	MUL	
guestsocialnetwork	NetworkID	int	NO	MUL	
guestsocialnetwork	ProfileURL	varchar	YES		
host	HostID	int	NO	PRI	auto_increment
host	GuestID	int	NO	MUL	
host	Rating	float	YES		
host	Verified	tinyint	YES		
host	HostSince	date	YES		
host	Stars	int	YES		
host	ExternalReviews	text	YES		
host	ReferredByHostID	int	YES	MUL	
location	LocationID	int	NO	PRI	auto_increment
location	CityID	int	YES	MUL	
location	Country	varchar	YES		
location	PartOfCity	varchar	YES		
location	Address	varchar	YES		
location	PhoneNumber	varchar	YES		
location	Email	varchar	YES		
loginhistory	LoginID	int	NO	PRI	auto_increment
loginhistory	GuestID	int	NO	MUL	
loginhistory	LoginTimestamp	datetime	YES		
loginhistory	IPAddress	varchar	YES		
notification	NotificationID	int	NO	PRI	auto_increment
notification	GuestID	int	NO	MUL	
notification	Content	text	YES		
notification	Timestamp	datetime	YES		
promotion	PromotionID	int	NO	PRI	auto_increment
promotion	VacationRentalID	int	NO	MUL	
promotion	DiscountPercentage	float	YES		
promotion	StartDate	date	YES		
promotion	EndDate	date	YES		
reservation	ReservationID	int	NO	PRI	auto_increment
reservation	BookingID	int	NO	MUL	
reservation	AdminID	int	NO	MUL	
reservation	DateOfReservation	date	YES		
reservation	PaymentStatus	enum	YES		
reservation	LengthOfStay	int	YES		
reservation	CancellationPolicy	text	YES		
reservation	RefundPolicy	text	YES		
review	ReviewID	int	NO	PRI	auto_increment
review	BookingID	int	NO	MUL	
review	ReviewerID	int	NO	MUL	
review	ReviewerType	enum	NO		
review	Rating	int	YES		
review	Comment	text	YES		

review	ReviewDate	date	YES		
room	RoomID	int	NO	PRI	auto_increment
room	VacationRentalID	int	NO	MUL	
room	RoomType	varchar	YES		
room	PricePerNight	float	YES		
room	AvailableFrom	date	YES		
room	AvailableTo	date	YES		
socialnetwork	NetworkID	int	NO	PRI	auto_increment
socialnetwork	NetworkName	varchar	YES		
socialnetwork	URL	varchar	YES		
transaction	TransactionID	int	NO	PRI	auto_increment
transaction	GuestID	int	NO	MUL	
transaction	BookingID	int	NO	MUL	
transaction	Amount	float	NO		
transaction	TransactionDate	datetime	NO		
transaction	PaymentMethod	enum	NO		
transaction	TransactionType	enum	NO		
transaction	RefundProcessedDate	datetime	YES		
transaction	Description	text	NO		
traveladmin	AdminID	int	NO	PRI	auto_increment
traveladmin	Name	varchar	NO		
traveladmin	Email	varchar	NO	UNI	
traveladmin	PhoneNumber	varchar	YES		
traveladmin	GuestID	int	NO	MUL	
vacationrental	VacationRentalID	int	NO	PRI	auto_increment
vacationrental	HostID	int	NO	MUL	
vacationrental	LocationID	int	NO	MUL	
vacationrental	PropertyType	varchar	YES		
vacationrental	Description	text	YES		
vacationrental	MaxGuests	int	YES		
vacationrental	RatePerPerson	float	YES		
vacationrental	OwnBathroom	tinyint	YES		
vacationrental	DogFriendly	tinyint	YES		
vacationrental	FreeParking	tinyint	YES		
vacationrental	NumberOfBeds	int	YES		
vacationrental	CalendarAvailability	text	YES		
vacationrental	ProximityToBeach	varchar	YES		
vacationrental	ProximityToShops	varchar	YES		
vacationrental	ProximityToSightSeeing	varchar	YES		
vacationrentalamenity	VacationRentalID	int	NO	PRI	
vacationrentalamenity	AmenityID	int	NO	PRI	
vacationrentalpolicy	VacationRentalID	int	NO	PRI	
vacationrentalpolicy	PolicyID	int	NO	PRI	

The Information Schema:

The information schema offers insights into how the database is organized, including its entities, foreign keys, and indexes.

Nr.	Name	Rows	Avg_row_length	Data_length	Index_length	Auto_increment
1	amenity	20	819	16384	0	21
2	booking	20	819	16384	32768	21
3	cancellationpolicy	20	819	16384	0	21
4	city	20	819	16384	0	21
5	customerservice	20	819	16384	16384	21
6	event	20	819	16384	16384	21
7	guest	20	819	16384	16384	21
8	guestsocialnetwork	20	819	16384	32768	21
9	host	20	819	16384	32768	21
10	location	20	819	16384	16384	21
11	loginhistory	20	819	16384	16384	21
12	notification	20	819	16384	16384	21
13	promotion	20	819	16384	16384	21
14	reservation	20	819	16384	32768	21
15	review	20	819	16384	32768	21
16	room	20	819	16384	16384	21
17	socialnetwork	20	819	16384	0	21
18	transaction	20	819	16384	32768	21
19	traveladmin	20	819	16384	32768	21
20	vacationrental	20	819	16384	32768	21
21	vacationrentalamenity	20	819	16384	16384	
22	vacationrentalpolicy	20	819	16384	16384	

Total Size of the Schema:

The total size of the database schema is approximately 0.75 MB, consisting of 0.34 MB of data and 0.41 MB of indexes. The 20 entries have an average size of approximately 0.02 to 0.05 MB, depending on the data and index requirements. An understanding of how much storage the database occupies is important for assessing its scalability.

Database	Data Size (MB)	Index Size (MB)	Total Size (MB)
airbnb_datamart_sky	0.34	0.41	0.75

Table Name	Number of Entries	Data Size (MB)	Index Size (MB)	Total Size (MB)
amenity	20	0.02	0.00	0.02
booking	20	0.02	0.03	0.05
cancellationpolicy	20	0.02	0.00	0.02
city	20	0.02	0.00	0.02
customerservice	20	0.02	0.02	0.03
event	20	0.02	0.02	0.03
guest	20	0.02	0.02	0.03
guestsocialnetwork	20	0.02	0.03	0.05
host	20	0.02	0.03	0.05
location	20	0.02	0.02	0.03
loginhistory	20	0.02	0.02	0.03
notification	20	0.02	0.02	0.03
promotion	20	0.02	0.02	0.03
reservation	20	0.02	0.03	0.05
review	20	0.02	0.03	0.05
room	20	0.02	0.02	0.03
socialnetwork	20	0.02	0.00	0.02
transaction	20	0.02	0.03	0.05
traveladmin	20	0.02	0.03	0.05
vacationrental	20	0.02	0.03	0.05
vacationrentalame...	20	0.02	0.02	0.03
vacationrentalpolicy	20	0.02	0.02	0.03

List of Foreign Key Constraints:

Foreign key constraints ensure that data in related tables stays consistent by linking records, preventing the creation of invalid entries and ensuring that related data is always properly connected.

Table Name	Constraint Name	Column Name	Referenced Table	Referenced Column
booking	booking_ibfk_1	GuestID	guest	GuestID
booking	booking_ibfk_2	RoomID	room	RoomID
customerservice	customerservice_ibfk_1	BookingID	booking	BookingID
event	event_ibfk_1	BookingID	booking	BookingID
guestsocialnetwork	guestsocialnetwork_ibfk_1	GuestID	guest	GuestID
guestsocialnetwork	guestsocialnetwork_ibfk_2	NetworkID	socialnetwork	NetworkID
host	host_ibfk_1	GuestID	guest	GuestID
host	host_ibfk_2	ReferredByHostID	host	HostID
location	location_ibfk_1	CityID	city	CityID
loginhistory	loginhistory_ibfk_1	GuestID	guest	GuestID
notification	notification_ibfk_1	GuestID	guest	GuestID
promotion	promotion_ibfk_1	VacationRentalID	vacationrental	VacationRentalID
reservation	reservation_ibfk_1	BookingID	booking	BookingID
reservation	reservation_ibfk_2	AdminID	traveladmin	AdminID
review	review_ibfk_1	BookingID	booking	BookingID
review	review_ibfk_2	ReviewerID	guest	GuestID
room	room_ibfk_1	VacationRentalID	vacationrental	VacationRentalID
transaction	transaction_ibfk_1	GuestID	guest	GuestID
transaction	transaction_ibfk_2	BookingID	booking	BookingID
traveladmin	traveladmin_ibfk_1	GuestID	guest	GuestID
vacationrental	vacationrental_ibfk_1	HostID	host	HostID
vacationrental	vacationrental_ibfk_2	LocationID	location	LocationID
vacationrentalame...	vacationrentalamenity_ib...	VacationRentalID	vacationrental	VacationRentalID
vacationrentalame...	vacationrentalamenity_ib...	AmenityID	amenity	AmenityID
vacationrentalpolicy	vacationrentalpolicy_ibfk_1	VacationRentalID	vacationrental	VacationRentalID
vacationrentalpolicy	vacationrentalpolicy_ibfk_2	PolicyID	cancellationpolicy	PolicyID

List of Tables and Indexes:

A list of tables in the schema, along with their corresponding indexes, will be included here. Indexes were added to optimize the performance of frequently joined columns, particularly foreign keys like GuestID, HostID, and BookingID. These indexes improve query performance when joining tables or filtering records, quickly locating information without scanning every record.

TABLE_NAME	INDEX_NAME	COLUMN_NAME	NON_UNIQUE	INDEX_TYPE
amenity	PRIMARY	AmenityID	0	BTREE
booking	GuestID	GuestID	1	BTREE
booking	PRIMARY	BookingID	0	BTREE
booking	RoomID	RoomID	1	BTREE
cancellationpolicy	PRIMARY	PolicyID	0	BTREE
city	PRIMARY	CityID	0	BTREE
customerservice	BookingID	BookingID	1	BTREE
customerservice	PRIMARY	CustomerServiceID	0	BTREE
event	BookingID	BookingID	1	BTREE
event	PRIMARY	EventID	0	BTREE
guest	Email	Email	0	BTREE
guest	PRIMARY	GuestID	0	BTREE
guestsocialnetwork	GuestID	GuestID	1	BTREE
guestsocialnetwork	NetworkID	NetworkID	1	BTREE
guestsocialnetwork	PRIMARY	GuestSocialNetwo...	0	BTREE
host	GuestID	GuestID	1	BTREE
host	PRIMARY	HostID	0	BTREE
host	ReferredBy...	ReferredByHostID	1	BTREE
location	CityID	CityID	1	BTREE
location	PRIMARY	LocationID	0	BTREE
loginhistory	GuestID	GuestID	1	BTREE
loginhistory	PRIMARY	LoginID	0	BTREE
notification	GuestID	GuestID	1	BTREE
notification	PRIMARY	NotificationID	0	BTREE
promotion	PRIMARY	PromotionID	0	BTREE
promotion	VacationRen...	VacationRentalID	1	BTREE
reservation	AdminID	AdminID	1	BTREE
reservation	BookingID	BookingID	1	BTREE
reservation	PRIMARY	ReservationID	0	BTREE
review	BookingID	BookingID	1	BTREE
review	PRIMARY	ReviewID	0	BTREE
review	ReviewerID	ReviewerID	1	BTREE
room	PRIMARY	RoomID	0	BTREE
room	VacationRen...	VacationRentalID	1	BTREE
socialnetwork	PRIMARY	NetworkID	0	BTREE
transaction	BookingID	BookingID	1	BTREE
transaction	GuestID	GuestID	1	BTREE
transaction	PRIMARY	TransactionID	0	BTREE
traveladmin	Email	Email	0	BTREE
traveladmin	GuestID	GuestID	1	BTREE
traveladmin	PRIMARY	AdminID	0	BTREE
vacationrental	HostID	HostID	1	BTREE
vacationrental	LocationID	LocationID	1	BTREE
vacationrental	PRIMARY	VacationRentalID	0	BTREE
vacationrentala...	AmenityID	AmenityID	1	BTREE
vacationrentala...	PRIMARY	VacationRentalID	0	BTREE
vacationrentala...	PRIMARY	AmenityID	0	BTREE
vacationrentalp...	PolicyID	PolicyID	1	BTREE
vacationrentalp...	PRIMARY	VacationRentalID	0	BTREE
vacationrentalp...	PRIMARY	PolicyID	0	BTREE

2. MySQL Installation Manual

By following these steps, the database can be set up and validated for use in managing an Airbnb-like platform.

Install MySQL and MySQL Workbench:

- Download and install MySQL Community Server (click [here](#)).
- MySQL Workbench for database management (click [here](#)).

Set Up Database Connection:

- Open MySQL Workbench after installation and set up a new database connection by entering your server details (host, username, and password).
- Create a new schema by clicking Create Schema in the toolbar, and name it (here: "airbnb_datamart_sky").

Import the SQL Dump File:

- Use the Data Import Tool located in the Server tab in MySQL Workbench.
- Select the dump.sql file from your directory.
- Import the file into the newly created schema. This will generate all tables, relationships, and data as defined in the SQL datafile.

Verify Database Structure:

- After importing, run SHOW TABLES to ensure all tables have been created.
- To check the data in any table, use queries such as SELECT * FROM Guest; to retrieve entries.
- You can also run metadata queries to validate the number of rows, indexes, and foreign key constraints.

Validation and Testing:

- Once the schema is set up, run queries to verify table relationships. Ensure all foreign key relationships are functioning correctly by joining tables such as Guest, Booking, and Transaction.

3. SQL-Datafile

```

-----
-- Airbnb Datamart Database Setup
-- Disabling foreign key checks for smooth table creation
-----
USE airbnb_datamart_sky;
SET FOREIGN_KEY_CHECKS = 0;

-----
-- Dropping existing tables (if they exist) to ensure clean setup
-----
-- Ordered to avoid foreign key constraint issues during drop
DROP TABLE IF EXISTS GuestSocialNetwork, SocialNetwork, Notification, Review, CustomerService,
Reservation,
    Transaction, Booking, Room, VacationRentalPolicy, CancellationPolicy, VacationRentalAmenity,
Amenity,
    VacationRental, Location, City, LoginHistory, TravelAdmin, Host, Guest, Event, Promotion;

-- Re-enable foreign key checks after dropping tables
SET FOREIGN_KEY_CHECKS = 1;

-----
-- Guest-related tables
-----
-- Guest table stores personal details of users who can book properties
CREATE TABLE Guest (
    GuestID INT AUTO_INCREMENT PRIMARY KEY,
    Name VARCHAR(255) NOT NULL,
    Email VARCHAR(255) NOT NULL UNIQUE,
    Password VARCHAR(255) NOT NULL,
    PhoneNumber VARCHAR(20),
    ProfilePicture VARCHAR(255),
    Street VARCHAR(255),
    City VARCHAR(255),
    State VARCHAR(255),
    Country VARCHAR(255),
    GDPRAcknowledgement BOOLEAN,
    LanguageSettings VARCHAR(255)
);

-- Host table stores details about property owners. Guests can also be hosts.
CREATE TABLE Host (
    HostID INT AUTO_INCREMENT PRIMARY KEY,
    GuestID INT NOT NULL,
    Rating FLOAT,
    Verified BOOLEAN,
    HostSince DATE,
    Stars INT,
    ExternalReviews TEXT,
    ReferredByHostID INT NULL,
    FOREIGN KEY (GuestID) REFERENCES Guest(GuestID) ON DELETE CASCADE, -- Cascade deletes to
remove related hosts
    FOREIGN KEY (ReferredByHostID) REFERENCES Host(HostID) ON DELETE SET NULL -- Prevent circular
references. Self-referencing (recursive) relationship
);

-- TravelAdmin table stores system admins who can manage bookings and policies
CREATE TABLE TravelAdmin (
    AdminID INT AUTO_INCREMENT PRIMARY KEY,
    Name VARCHAR(255) NOT NULL,
    Email VARCHAR(255) NOT NULL UNIQUE,

```

```

        PhoneNumber VARCHAR(20),
        GuestID INT NOT NULL,
        FOREIGN KEY (GuestID) REFERENCES Guest(GuestID)
    );

-----
-- Social network and guest login activity
-----
-- SocialNetwork table stores social media networks that guests can link to
CREATE TABLE SocialNetwork (
    NetworkID INT AUTO_INCREMENT PRIMARY KEY,
    NetworkName VARCHAR(100),
    URL VARCHAR(255)
);

-- GuestSocialNetwork table links guests to their social media profiles
CREATE TABLE GuestSocialNetwork (
    GuestSocialNetworkID INT AUTO_INCREMENT PRIMARY KEY,
    GuestID INT NOT NULL,
    NetworkID INT NOT NULL,
    ProfileURL VARCHAR(255),
    FOREIGN KEY (GuestID) REFERENCES Guest(GuestID),
    FOREIGN KEY (NetworkID) REFERENCES SocialNetwork(NetworkID)
);

-- LoginHistory logs guest login details for security purposes
CREATE TABLE LoginHistory (
    LoginID INT AUTO_INCREMENT PRIMARY KEY,
    GuestID INT NOT NULL,
    LoginTimestamp DATETIME,
    IPAddress VARCHAR(45),
    FOREIGN KEY (GuestID) REFERENCES Guest(GuestID)
);

-- Notification table for system alerts and messages to guests
CREATE TABLE Notification (
    NotificationID INT AUTO_INCREMENT PRIMARY KEY,
    GuestID INT NOT NULL,
    Content TEXT,
    Timestamp DATETIME,
    FOREIGN KEY (GuestID) REFERENCES Guest(GuestID)
);

-----
-- Location and rental-related tables
-----
-- City table stores city details, referenced by properties
CREATE TABLE City (
    CityID INT AUTO_INCREMENT PRIMARY KEY,
    CityName VARCHAR(100),
    Country VARCHAR(100)
);

-- Location table links specific addresses to cities
CREATE TABLE Location (
    LocationID INT AUTO_INCREMENT PRIMARY KEY,
    CityID INT,
    Country VARCHAR(100),
    PartOfCity VARCHAR(100),
    Address VARCHAR(255),
    PhoneNumber VARCHAR(20),
    Email VARCHAR(255),
    FOREIGN KEY (CityID) REFERENCES City(CityID)
);

```

```

-- VacationRental table stores rental property details
CREATE TABLE VacationRental (
    VacationRentalID INT AUTO_INCREMENT PRIMARY KEY,
    HostID INT NOT NULL,
    LocationID INT NOT NULL,
    PropertyType VARCHAR(100),
    Description TEXT,
    MaxGuests INT,
    RatePerPerson FLOAT,
    OwnBathroom BOOLEAN,
    DogFriendly BOOLEAN,
    FreeParking BOOLEAN,
    NumberOfBeds INT,
    CalendarAvailability TEXT,
    ProximityToBeach VARCHAR(255),
    ProximityToShops VARCHAR(255),
    ProximityToSightSeeing VARCHAR(255),
    FOREIGN KEY (HostID) REFERENCES Host(HostID) ON DELETE CASCADE,
    FOREIGN KEY (LocationID) REFERENCES Location(LocationID) ON DELETE CASCADE
);

-----
-- Room, Amenity, and Policy-related tables
-----
-- Room table links specific rooms to rentals with availability and pricing
CREATE TABLE Room (
    RoomID INT AUTO_INCREMENT PRIMARY KEY,
    VacationRentalID INT NOT NULL,
    RoomType VARCHAR(100),
    PricePerNight FLOAT,
    AvailableFrom DATE,
    AvailableTo DATE,
    FOREIGN KEY (VacationRentalID) REFERENCES VacationRental(VacationRentalID)
);

-- Amenity table stores general amenities like Wi-Fi, parking, etc.
CREATE TABLE Amenity (
    AmenityID INT AUTO_INCREMENT PRIMARY KEY,
    AmenityName VARCHAR(100),
    Description TEXT
);

-- Linking amenities to vacation rentals
CREATE TABLE VacationRentalAmenity (
    VacationRentalID INT NOT NULL,
    AmenityID INT NOT NULL,
    PRIMARY KEY (VacationRentalID, AmenityID),
    FOREIGN KEY (VacationRentalID) REFERENCES VacationRental(VacationRentalID),
    FOREIGN KEY (AmenityID) REFERENCES Amenity(AmenityID)
);

-- Cancellation policies for rentals
CREATE TABLE CancellationPolicy (
    PolicyID INT AUTO_INCREMENT PRIMARY KEY,
    PolicyName VARCHAR(100),
    Description TEXT
);

-- Linking cancellation policies to rentals
CREATE TABLE VacationRentalPolicy (
    VacationRentalID INT NOT NULL,
    PolicyID INT NOT NULL,
    PRIMARY KEY (VacationRentalID, PolicyID),

```

```

        FOREIGN KEY (VacationRentalID) REFERENCES VacationRental(VacationRentalID),
        FOREIGN KEY (PolicyID) REFERENCES CancellationPolicy(PolicyID)
    );

-----
-- Booking-related tables
-----
-- Booking table stores details about guest bookings
CREATE TABLE Booking (
    BookingID INT AUTO_INCREMENT PRIMARY KEY,
    GuestID INT NOT NULL,
    RoomID INT NOT NULL,
    BookingDate DATETIME NOT NULL,
    CheckInDate DATE NOT NULL,
    CheckOutDate DATE NOT NULL,
    TotalPrice FLOAT NOT NULL,
    PaymentStatus ENUM('Paid', 'Pending', 'Cancelled') NOT NULL,
    LengthOfStay INT NOT NULL,
    CancellationDeadline DATE,
    CancellationRefund FLOAT,
    DateOfCancellation DATE,
    HostPayout FLOAT,
    EventName VARCHAR(255),
    StartDate DATE,
    EndDate DATE,
    Description TEXT,
    FOREIGN KEY (GuestID) REFERENCES Guest(GuestID) ON DELETE CASCADE,
    FOREIGN KEY (RoomID) REFERENCES Room(RoomID) ON DELETE CASCADE
);

-----
-- Transaction, Reservation, and Review tables
-----
-- Storing transactions for payments and refunds
CREATE TABLE Transaction (
    TransactionID INT AUTO_INCREMENT PRIMARY KEY,
    GuestID INT NOT NULL,
    BookingID INT NOT NULL,
    Amount FLOAT NOT NULL,
    TransactionDate DATETIME NOT NULL,
    PaymentMethod ENUM('CreditCard', 'BankTransfer', 'Cash', 'Voucher', 'PayPal', 'ApplePay',
'GPay', 'CreditCard_MasterCard', 'CreditCard_Visa',
'CreditCard_AMEX', 'CreditCard_FirstCard', 'CreditCard_DinersClub', 'Maestro',
'SOFORT_Payment', 'BNPL', 'Klarna') NOT NULL,
    TransactionType ENUM('Payment', 'Refund') NOT NULL,
    RefundProcessedDate DATETIME NULL,
    Description TEXT NOT NULL,
    FOREIGN KEY (GuestID) REFERENCES Guest(GuestID),
    FOREIGN KEY (BookingID) REFERENCES Booking(BookingID)
);

-- Reservation table storing additional booking details and policies
CREATE TABLE Reservation (
    ReservationID INT AUTO_INCREMENT PRIMARY KEY,
    BookingID INT NOT NULL,
    AdminID INT NOT NULL,
    DateOfReservation DATE,
    PaymentStatus ENUM('Paid', 'Pending', 'Cancelled'),
    LengthOfStay INT,
    CancellationPolicy TEXT,
    RefundPolicy TEXT,
    FOREIGN KEY (BookingID) REFERENCES Booking(BookingID),
    FOREIGN KEY (AdminID) REFERENCES TravelAdmin(AdminID)
);

```

```

-- Review table for collecting feedback from guests or hosts
CREATE TABLE Review (
    ReviewID INT AUTO_INCREMENT PRIMARY KEY,
    BookingID INT NOT NULL,
    ReviewerID INT NOT NULL,
    ReviewerType ENUM('Guest', 'Host') NOT NULL,
    Rating INT,
    Comment TEXT,
    ReviewDate DATE,
    FOREIGN KEY (BookingID) REFERENCES Booking(BookingID) ON DELETE CASCADE,
    FOREIGN KEY (ReviewerID) REFERENCES Guest(GuestID) ON DELETE CASCADE
);

-----
-- Customer service and event management
-----

-- Customer service table for managing support tickets related to bookings
CREATE TABLE CustomerService (
    CustomerServiceID INT AUTO_INCREMENT PRIMARY KEY,
    BookingID INT NOT NULL,
    IssueDescription TEXT,
    Resolution TEXT,
    ContactMethod VARCHAR(255),
    ResolutionDate DATE,
    FOREIGN KEY (BookingID) REFERENCES Booking(BookingID)
);

-- Event table to link events to bookings, such as local activities
CREATE TABLE Event (
    EventID INT AUTO_INCREMENT PRIMARY KEY,
    BookingID INT NOT NULL,
    EventName VARCHAR(255),
    StartDate DATE,
    EndDate DATE,
    Description TEXT,
    FOREIGN KEY (BookingID) REFERENCES Booking(BookingID)
);

-----
-- Promotion table for managing rental discounts
-----

CREATE TABLE Promotion (
    PromotionID INT AUTO_INCREMENT PRIMARY KEY,
    VacationRentalID INT NOT NULL,
    DiscountPercentage FLOAT,
    StartDate DATE,
    EndDate DATE,
    FOREIGN KEY (VacationRentalID) REFERENCES VacationRental(VacationRentalID)
);

-----
-- Insert test data into the Guest table
-- Guest data should cover various use cases such as international guests, language preferences,
etc.
-----

INSERT INTO Guest (Name, Email, Password, PhoneNumber, ProfilePicture, Street, City, State,
Country, GDPRAcknowledgement, LanguageSettings)
VALUES
    ('Hans Müller', 'hans.mueller@example.de', 'password111', '491523456789', 'hans_profile.jpg',
'123 Main St', 'Berlin', '', 'Germany', TRUE, 'German'),
    ('Emma Karlsson', 'emma.karlsson@example.se', 'password112', '467021234567',
'emma_profile.jpg', '456 Elm St', 'Stockholm', '', 'Sweden', TRUE, 'Swedish'),

```

```
(
    ('Jean Dupont', 'jean.dupont@example.fr', 'password113', '33122334455', 'jean_profile.jpg',
    '789 Oak St', 'Paris', '', 'France', TRUE, 'French'),
    ('Giovanni Rossi', 'giovanni.rossi@example.it', 'password211', '390123456789',
    'giovanni_profile.jpg', '101 Maple St', 'Rome', '', 'Italy', TRUE, 'Italian'),
    ('Seo-jun Lee', 'seo.jun.lee@example.kr', 'password212', '821012345678', 'seo_profile.jpg',
    '202 Pine St', 'Seoul', '', 'South Korea', TRUE, 'Korean'),
    ('Sakura Tanaka', 'sakura.tanaka@example.jp', 'password213', '819012345678',
    'sakura_profile.jpg', '303 Birch St', 'Tokyo', '', 'Japan', TRUE, 'Japanese'),
    ('Anna Svensson', 'anna.svensson@example.se', 'password100', '46709876543',
    'anna_profile.jpg', '404 Cedar St', 'Gothenburg', '', 'Sweden', TRUE, 'Swedish'),
    ('Friedrich Becker', 'friedrich.becker@example.de', 'password101', '4915123456789',
    'friedrich_profile.jpg', '505 Pine St', 'Munich', '', 'Germany', TRUE, 'German'),
    ('Pierre Martin', 'pierre.martin@example.fr', 'password110', '33122334455',
    'pierre_profile.jpg', '606 Maple St', 'Lyon', '', 'France', TRUE, 'French'),
    ('Alessandro Bianchi', 'alessandro.bianchi@example.it', 'password200', '390223344556',
    'alessandro_profile.jpg', '707 Oak St', 'Milan', '', 'Italy', TRUE, 'Italian'),
    ('Ahmed El-Sayed', 'ahmed.el-sayed@example.eg', 'password202', '201234567890',
    'ahmed_profile.jpg', '808 Pine St', 'Cairo', '', 'Egypt', TRUE, 'Arabic'),
    ('John Smith', 'john.smith@example.co.uk', 'password201', '441234567890', 'john_profile.jpg',
    '909 Birch St', 'London', '', 'United Kingdom', TRUE, 'English (UK)'),
    ('Olivia Brown', 'olivia.brown@example.au', 'password220', '611234567890',
    'olivia_profile.jpg', '1010 Cedar St', 'Sydney', '', 'Australia', TRUE, 'English (Australia)'),
    ('Michael Johnson', 'michael.johnson@example.us', 'password203', '11234567890',
    'michael_profile.jpg', '1111 Maple St', 'New York', 'NY', 'USA', TRUE, 'English (USA)'),
    ('Emma Williams', 'emma.williams@example.ca', 'password103', '11234567891',
    'emma_profile.jpg', '1212 Oak St', 'Toronto', '', 'Canada', TRUE, 'English (Canada)'),
    ('Chloe Dubois', 'chloe.dubois@example.ch', 'password301', '33122334456',
    'chloe_profile.jpg', '1313 Pine St', 'Geneva', '', 'Switzerland', TRUE, 'French (Switzerland)'),
    ('Mats Andersson', 'mats.andersson@example.se', 'password300', '46709876544',
    'mats_profile.jpg', '1414 Cedar St', 'Malmö', '', 'Sweden', TRUE, 'Swedish'),
    ('Sophie Schmidt', 'sophie.schmidt@example.de', 'password302', '491523456789',
    'sophie_profile.jpg', '1515 Maple St', 'Hamburg', '', 'Germany', TRUE, 'German'),
    ('Luca Bruni', 'luca.bruni@example.it', 'password311', '39034567890', 'luca_profile.jpg',
    '1616 Oak St', 'Florence', '', 'Italy', TRUE, 'Italian'),
    ('Isabelle Laurent', 'isabelle.laurent@example.fr', 'password321', '33122334455',
    'isabelle_profile.jpg', '1717 Pine St', 'Nice', '', 'France', TRUE, 'French'));
```

```
-----
-- Insert test data into the Host table
-- This links Hosts to Guests, making some Guests also Hosts
-----
```

```
INSERT INTO Host (GuestID, Rating, Verified, HostSince, Stars, ExternalReviews, ReferredByHostID)
VALUES
```

```
(1, 4.5, TRUE, '2020-01-01', 5, 'Great reviews from external sites.', NULL),
(2, 4.0, TRUE, '2021-03-15', 4, 'Positive feedback from external platforms.', NULL),
(3, 4.8, TRUE, '2019-11-20', 5, 'Outstanding reviews.', NULL),
(4, 4.2, TRUE, '2020-06-25', 4, 'Excellent ratings on various platforms.', NULL),
(5, 4.7, TRUE, '2021-07-10', 5, 'Highly recommended by guests.', NULL),
(6, 4.9, TRUE, '2021-05-22', 5, 'Top-rated host with many positive reviews.', NULL),
(7, 4.3, TRUE, '2019-12-05', 4, 'Good reviews overall.', NULL),
(8, 4.6, TRUE, '2020-02-15', 5, 'Consistent positive feedback from guests.', NULL),
(9, 4.1, TRUE, '2021-09-01', 4, 'Mixed reviews but generally positive.', NULL),
(10, 4.4, TRUE, '2021-08-18', 5, 'Great experience reported by guests.', NULL),
(11, 4.3, TRUE, '2020-10-10', 4, 'Strong feedback from external reviews.', NULL),
(12, 4.8, TRUE, '2021-01-01', 5, 'Highly recommended by several sources.', NULL),
(13, 4.7, TRUE, '2020-03-20', 5, 'Great host with excellent reviews.', NULL),
(14, 4.9, TRUE, '2019-07-07', 5, 'One of the top-rated hosts.', NULL),
(15, 4.5, TRUE, '2021-04-15', 5, 'Guests consistently leave positive feedback.', NULL),
(16, 4.6, TRUE, '2019-09-25', 5, 'Highly rated by multiple sources.', NULL),
(17, 4.2, TRUE, '2020-11-10', 4, 'Reviews indicate a good experience.', NULL),
(18, 4.5, TRUE, '2021-02-22', 5, 'Popular host with lots of positive reviews.', NULL),
(19, 4.7, TRUE, '2020-12-15', 5, 'Guests frequently recommend this host.', NULL),
(20, 4.4, TRUE, '2021-03-10', 4, 'Well-reviewed host with solid ratings.', NULL);
```

```

-----
-- Insert test data into the SocialNetwork table
-- Define different social media platforms
-----

```

```

INSERT INTO SocialNetwork (NetworkName, URL)
VALUES
    ('Facebook', 'https://www.facebook.com'),
    ('Twitter', 'https://www.twitter.com'),
    ('Instagram', 'https://www.instagram.com'),
    ('LinkedIn', 'https://www.linkedin.com'),
    ('WhatsApp', 'https://www.whatsapp.com'),
    ('WeChat', 'https://www.wechat.com'),
    ('VK', 'https://www.vk.com'),
    ('Snapchat', 'https://www.snapchat.com'),
    ('TikTok', 'https://www.tiktok.com'),
    ('Pinterest', 'https://www.pinterest.com'),
    ('Reddit', 'https://www.reddit.com'),
    ('Tumblr', 'https://www.tumblr.com'),
    ('Flickr', 'https://www.flickr.com'),
    ('YouTube', 'https://www.youtube.com'),
    ('Vimeo', 'https://www.vimeo.com'),
    ('Discord', 'https://www.discord.com'),
    ('Telegram', 'https://www.telegram.org'),
    ('Signal', 'https://www.signal.org'),
    ('Baidu Tieba', 'https://tieba.baidu.com'),
    ('Douban', 'https://www.douban.com');

```

```

-----
-- Insert test data into the GuestSocialNetwork table
-- Social media profiles linked to guests
-----

```

```

INSERT INTO GuestSocialNetwork (GuestID, NetworkID, ProfileURL)
VALUES
    (1, 1, 'https://www.facebook.com/hans.mueller'),
    (2, 2, 'https://www.twitter.com/emma.karlsson'),
    (3, 3, 'https://www.instagram.com/jean.dupont'),
    (4, 4, 'https://www.linkedin.com/in/giovanni.rossi'),
    (5, 5, 'https://www.whatsapp.com/seo.jun.lee'),
    (6, 6, 'https://www.wechat.com/sakura.tanaka'),
    (7, 7, 'https://www.vk.com/anna.svensson'),
    (8, 8, 'https://www.snapchat.com/friedrich.becker'),
    (9, 9, 'https://www.tiktok.com/pierre.martin'),
    (10, 10, 'https://www.pinterest.com/alessandro.bianchi'),
    (11, 11, 'https://www.reddit.com/user/ahmed.elsayed'),
    (12, 12, 'https://www.tumblr.com/john.smith'),
    (13, 13, 'https://www.flickr.com/olivia.brown'),
    (14, 14, 'https://www.youtube.com/michael.johnson'),
    (15, 15, 'https://www.vimeo.com/emma.williams'),
    (16, 16, 'https://www.discord.com/chloe.dubois'),
    (17, 17, 'https://www.telegram.org/mats.andersson'),
    (18, 18, 'https://www.signal.org/sophie.schmidt'),
    (19, 19, 'https://tieba.baidu.com/luca.bruni'),
    (20, 20, 'https://www.douban.com/isabelle.laurent');

```

```

-----
-- Insert test data into the Notification table
-- Notifications sent to guests
-----

```

```

INSERT INTO Notification (GuestID, Content, Timestamp)
VALUES
    (1, 'Welcome to our service!', '2024-09-01 08:10:00'),
    (2, 'Your booking has been confirmed.', '2024-09-01 09:15:00'),
    (3, 'Payment successful.', '2024-09-01 10:20:00'),

```

```
(4, 'Your stay in Rome is coming up.', '2024-09-01 11:25:00'),
(5, 'Check-in available for your booking.', '2024-09-01 12:30:00'),
(6, 'Reminder: Leave a review for your recent stay.', '2024-09-01 13:35:00'),
(7, 'New message from your host.', '2024-09-01 14:40:00'),
(8, 'Special offer for your next stay.', '2024-09-01 15:45:00'),
(9, 'Booking cancellation successful.', '2024-09-01 16:50:00'),
(10, 'Your refund has been processed.', '2024-09-01 17:55:00'),
(11, 'New property added in your favorite city.', '2024-09-01 18:00:00'),
(12, 'Update your profile to get better recommendations.', '2024-09-01 19:05:00'),
(13, 'New loyalty points added to your account.', '2024-09-01 20:10:00'),
(14, 'Last chance to book your dream vacation.', '2024-09-01 21:15:00'),
(15, 'Your stay in New York is coming up.', '2024-09-01 22:20:00'),
(16, 'Your booking for Zurich has been confirmed.', '2024-09-01 23:25:00'),
(17, 'Check-out reminder for your stay.', '2024-09-02 00:30:00'),
(18, 'Thank you for staying with us!', '2024-09-02 01:35:00'),
(19, 'Special promotion: 20% off your next booking.', '2024-09-02 02:40:00'),
(20, 'Your profile has been updated successfully.', '2024-09-02 03:45:00');
```

```
-----
-- Insert test data into the LoginHistory table
-- Track guest login activity with timestamps and IP addresses
-----
```

```
INSERT INTO LoginHistory (GuestID, LoginTimestamp, IPAddress)
VALUES
(1, '2024-09-01 08:00:00', '192.168.1.1'),
(2, '2024-09-01 09:00:00', '192.168.1.2'),
(3, '2024-09-01 10:00:00', '192.168.1.3'),
(4, '2024-09-01 11:00:00', '192.168.1.4'),
(5, '2024-09-01 12:00:00', '192.168.1.5'),
(6, '2024-09-01 13:00:00', '192.168.1.6'),
(7, '2024-09-01 14:00:00', '192.168.1.7'),
(8, '2024-09-01 15:00:00', '192.168.1.8'),
(9, '2024-09-01 16:00:00', '192.168.1.9'),
(10, '2024-09-01 17:00:00', '192.168.1.10'),
(11, '2024-09-01 18:00:00', '192.168.1.11'),
(12, '2024-09-01 19:00:00', '192.168.1.12'),
(13, '2024-09-01 20:00:00', '192.168.1.13'),
(14, '2024-09-01 21:00:00', '192.168.1.14'),
(15, '2024-09-01 22:00:00', '192.168.1.15'),
(16, '2024-09-01 23:00:00', '192.168.1.16'),
(17, '2024-09-02 00:00:00', '192.168.1.17'),
(18, '2024-09-02 01:00:00', '192.168.1.18'),
(19, '2024-09-02 02:00:00', '192.168.1.19'),
(20, '2024-09-02 03:00:00', '192.168.1.20');
```

```
-----
-- Insert test data into the City table
-- Define cities and their corresponding countries
-----
```

```
INSERT INTO City (CityName, Country)
VALUES
('Berlin', 'Germany'),
('Stockholm', 'Sweden'),
('Paris', 'France'),
('Rome', 'Italy'),
('Seoul', 'South Korea'),
('Tokyo', 'Japan'),
('Gothenburg', 'Sweden'),
('Munich', 'Germany'),
('Lyon', 'France'),
('Milan', 'Italy'),
('Cairo', 'Egypt'),
('London', 'United Kingdom'),
('Sydney', 'Australia');
```

```

('New York', 'USA'),
('Toronto', 'Canada'),
('Zurich', 'Switzerland'),
('Malmo', 'Sweden'),
('Hamburg', 'Germany'),
('Florence', 'Italy'),
('Nice', 'France');

-----
-- Insert test data into the Location table
-- Specify addresses and locations for vacation rentals within cities
-----
INSERT INTO Location (CityID, Country, PartOfCity, Address, PhoneNumber, Email)
VALUES
    (1, 'Germany', 'Mitte', 'Alexanderplatz 1, 10178 Berlin', '49123456789', 'info@berlin.de'),
    (2, 'Sweden', 'Södermalm', 'Götgatan 100, 11862 Stockholm', '46701234567',
'info@stockholm.se'),
    (3, 'France', 'Montmartre', 'Rue Lepic 18, 75018 Paris', '33123456789', 'info@paris.fr'),
    (4, 'Italy', 'Trastevere', 'Via della Scala 12, 00153 Rome', '390123456789', 'info@rome.it'),
    (5, 'South Korea', 'Gangnam-gu', 'Teheran-ro 100, 06134 Seoul', '821012345678',
'info@seoul.kr'),
    (6, 'Japan', 'Shibuya', 'Harajuku 1-23-45, 150-0001 Tokyo', '819012345678', 'info@tokyo.jp'),
    (7, 'Sweden', 'Vasastan', 'Sveavägen 50, 11359 Gothenburg', '46709876543',
'info@gothenburg.se'),
    (8, 'Germany', 'Altstadt-Lehel', 'Marienplatz 1, 80331 Munich', '4915123456789',
'info@munich.de'),
    (9, 'France', 'La Croix-Rousse', 'Rue de la République 50, 69001 Lyon', '33112233445',
'info@lyon.fr'),
    (10, 'Italy', 'Brera', 'Via Fiori Chiari 20, 20121 Milan', '390223344556', 'info@milan.it'),
    (11, 'Egypt', 'Zamalek', '26th of July St, Cairo', '201234567890', 'info@cairo.eg'),
    (12, 'United Kingdom', 'Westminster', '10 Downing St, London SW1A 2AA', '441234567890',
'info@london.co.uk'),
    (13, 'Australia', 'The Rocks', 'George St 140, 2000 Sydney', '611234567890',
'info@sydney.au'),
    (14, 'USA', 'Manhattan', '5th Avenue 350, New York, NY 10001', '11234567890',
'info@newyork.us'),
    (15, 'Canada', 'Downtown', 'Bay St 200, Toronto, ON M5J 2R8', '11234567891',
'info@toronto.ca'),
    (16, 'Switzerland', 'Altstadt', 'Bahnhofstrasse 1, 8001 Zurich', '41234567890',
'info@zurich.ch'),
    (17, 'Sweden', 'Gamla Staden', 'Södra Förstadsgatan 15, 21143 Malmo', '46709876544',
'info@malmo.se'),
    (18, 'Germany', 'HafenCity', 'Sandtorkai 30, 20457 Hamburg', '4915223456789',
'info@hamburg.de'),
    (19, 'Italy', 'Santa Maria Novella', 'Piazza di Santa Maria Novella, 50123 Florence',
'390334455667', 'info@florence.it'),
    (20, 'France', 'Promenade des Anglais', 'Quai des États-Unis 50, 06300 Nice', '33122334455',
'info@nice.fr');

-----
-- Insert test data into the VacationRental table
-- Define vacation rentals with relevant details like host, location, and property features
-----
INSERT INTO VacationRental (
    HostID, LocationID, PropertyType, Description, MaxGuests, RatePerPerson, OwnBathroom,
    DogFriendly, FreeParking, NumberOfBeds, CalendarAvailability, ProximityToBeach,
    ProximityToShops, ProximityToSightSeeing)
VALUES
    (1, 1, 'Apartment', 'Beautiful luxury apartment in the city center.', 4, 120.00, 1, 1, 1, 2,
'Available', '500m', '200m', '300m'),
    (2, 2, 'House', 'Spacious family house with a garden.', 6, 150.00, 1, 1, 1, 3, 'Available',
'1km', '500m', '700m'),
    (3, 3, 'Apartment', 'Cozy apartment near the river.', 2, 80.00, 1, 0, 1, 1, 'Available', '300m',
'100m', '200m'),

```

```
(4, 4, 'Villa', 'Luxury villa with ocean views.', 8, 250.00, 1, 1, 1, 4, 'Available', '50m',
'1km', '500m'),
(5, 5, 'Condo', 'Modern condo in a high-rise building.', 3, 90.00, 1, 0, 1, 1, 'Available',
'800m', '200m', '400m'),
(6, 6, 'Cottage', 'Charming cottage in the countryside.', 5, 130.00, 1, 1, 1, 2, 'Available',
'10km', '5km', '6km'),
(7, 7, 'Townhouse', 'Stylish townhouse in a vibrant neighborhood.', 4, 140.00, 1, 1, 1, 2,
'Available', '1km', '500m', '800m'),
(8, 8, 'Penthouse', 'Top-floor penthouse with panoramic city views.', 4, 300.00, 1, 1, 1, 2,
'Available', '200m', '100m', '300m'),
(9, 9, 'Villa', 'Exclusive villa with private pool.', 10, 400.00, 1, 1, 1, 5, 'Available', '50m',
'300m', '400m'),
(10, 10, 'Apartment', 'Affordable apartment close to the city center.', 3, 70.00, 1, 0, 1, 1,
'Available', '600m', '400m', '500m'),
(11, 11, 'Loft', 'Stylish loft in the heart of the city.', 2, 150.00, 1, 0, 1, 1, 'Available',
'400m', '150m', '300m'),
(12, 12, 'Chalet', 'Cozy mountain chalet with fireplace.', 6, 180.00, 1, 1, 1, 3, 'Available',
'1km', '600m', '800m'),
(13, 13, 'Bungalow', 'Beachfront bungalow in a tropical setting.', 4, 200.00, 1, 1, 1, 2,
'Available', '50m', '100m', '200m'),
(14, 14, 'Cabin', 'Secluded cabin in the woods.', 5, 120.00, 1, 1, 1, 2, 'Available', '5km',
'3km', '4km'),
(15, 15, 'Studio', 'Compact studio in a quiet area.', 2, 60.00, 1, 0, 1, 1, 'Available', '500m',
'300m', '400m'),
(16, 16, 'Mansion', 'Luxurious mansion with sprawling gardens.', 12, 500.00, 1, 1, 1, 6,
'Available', '1km', '500m', '700m'),
(17, 17, 'Townhouse', 'Spacious townhouse perfect for families.', 6, 160.00, 1, 1, 1, 3,
'Available', '700m', '200m', '300m'),
(18, 18, 'Villa', 'Exclusive villa with private beach access.', 8, 350.00, 1, 1, 1, 4,
'Available', '0m', '100m', '200m'),
(19, 19, 'Apartment', 'Cozy apartment for budget travelers.', 2, 50.00, 1, 0, 1, 1, 'Available',
'800m', '400m', '600m'),
(20, 20, 'Cottage', 'Charming cottage with garden.', 5, 140.00, 1, 1, 1, 2, 'Available', '2km',
'500m', '700m');
```

```
-----
-- Insert test data into the Room table
-- Room availability and pricing for specific rentals
-----
```

```
INSERT INTO Room (VacationRentalID, RoomType, PricePerNight, AvailableFrom, AvailableTo)
VALUES
```

```
(1, 'Deluxe', 150.00, '2024-01-01', '2024-03-31'),
(2, 'Standard', 120.00, '2024-04-01', '2024-06-30'),
(3, 'Deluxe', 180.00, '2024-07-01', '2024-09-30'),
(4, 'Suite', 250.00, '2024-10-01', '2024-12-31'),
(5, 'Standard', 90.00, '2024-02-01', '2024-05-31'),
(6, 'Deluxe', 140.00, '2024-06-01', '2024-09-30'),
(7, 'Cottage', 130.00, '2024-04-01', '2024-08-31'),
(8, 'Standard', 100.00, '2024-05-01', '2024-10-31'),
(9, 'Deluxe', 160.00, '2024-03-01', '2024-06-30'),
(10, 'Standard', 120.00, '2024-07-01', '2024-12-31'),
(11, 'Economy', 85.00, '2024-01-15', '2024-04-15'),
(12, 'Standard', 200.00, '2024-05-01', '2024-08-31'),
(13, 'Penthouse', 300.00, '2024-06-01', '2024-11-30'),
(14, 'Loft', 160.00, '2024-03-01', '2024-07-31'),
(15, 'Economy', 100.00, '2024-02-01', '2024-05-31'),
(16, 'Chalet', 220.00, '2024-06-01', '2024-09-30'),
(17, 'Townhouse', 130.00, '2024-04-01', '2024-07-31'),
(18, 'Studio', 70.00, '2024-03-01', '2024-10-31'),
(19, 'Suite', 240.00, '2024-05-01', '2024-09-30'),
(20, 'Standard', 105.00, '2024-07-01', '2024-12-31');
```

```
-----
-- Insert test data into the Amenity table
```

```
-- Defining various amenities available at vacation rentals
```

```
-----
INSERT INTO Amenity (AmenityName, Description)
VALUES
    ('WiFi', 'High-speed internet access'),
    ('Air Conditioning', 'Room equipped with air conditioning'),
    ('Heating', 'Central heating system'),
    ('Parking', 'Free parking space available'),
    ('Swimming Pool', 'Access to a swimming pool'),
    ('Kitchen', 'Fully equipped kitchen'),
    ('Laundry', 'Laundry facilities available'),
    ('Gym', 'Access to a gym'),
    ('Pet Friendly', 'Pets are allowed'),
    ('Breakfast', 'Breakfast included in the stay'),
    ('Balcony', 'Room with a balcony'),
    ('Terrace', 'Access to a terrace'),
    ('Sea View', 'Room with a view of the sea'),
    ('Fireplace', 'Room equipped with a fireplace'),
    ('TV', 'Television available in the room'),
    ('Garden', 'Access to a garden'),
    ('Hot Tub', 'Hot tub available'),
    ('Sauna', 'Access to a sauna'),
    ('Barbecue', 'Barbecue facilities available'),
    ('Bicycle Rental', 'Bicycles available for rent');
```

```
-----
-- Insert test data into the VacationRentalAmenity table
-- Associating amenities with vacation rentals
-----
```

```
INSERT INTO VacationRentalAmenity (VacationRentalID, AmenityID)
VALUES
    (1, 1), -- WiFi for VacationRentalID 1
    (2, 2), -- Air Conditioning for VacationRentalID 2
    (3, 3), -- Heating for VacationRentalID 3
    (4, 4), -- Parking for VacationRentalID 4
    (5, 5), -- Swimming Pool for VacationRentalID 5
    (6, 6), -- Kitchen for VacationRentalID 6
    (7, 7), -- Laundry for VacationRentalID 7
    (8, 8), -- Gym for VacationRentalID 8
    (9, 9), -- Pet Friendly for VacationRentalID 9
    (10, 10), -- Breakfast for VacationRentalID 10
    (11, 11), -- Balcony for VacationRentalID 11
    (12, 12), -- Terrace for VacationRentalID 12
    (13, 13), -- Sea View for VacationRentalID 13
    (14, 14), -- Fireplace for VacationRentalID 14
    (15, 15), -- TV for VacationRentalID 15
    (16, 16), -- Garden for VacationRentalID 16
    (17, 17), -- Hot Tub for VacationRentalID 17
    (18, 18), -- Sauna for VacationRentalID 18
    (19, 19), -- Barbecue for VacationRentalID 19
    (20, 20); -- Bicycle Rental for VacationRentalID 20
```

```
-----
-- Insert test data into the Booking table
-- Bookings link Guests to Rooms and store critical booking details
-----
```

```
INSERT INTO Booking (GuestID, RoomID, BookingDate, CheckInDate, CheckOutDate, TotalPrice,
PaymentStatus, LengthOfStay, CancellationDeadline, CancellationRefund, DateOfCancellation,
HostPayout, EventName, StartDate, EndDate, Description)
VALUES
    (1, 1, NOW(), '2024-09-01', '2024-09-07', 900.00, 'Pending', 6, '2024-08-29', 800.00, NULL,
850.00, 'Berlin Marathon', '2024-09-01', '2024-09-07', 'Running event in Berlin'),
    (2, 2, NOW(), '2024-09-05', '2024-09-10', 600.00, 'Paid', 5, '2024-09-03', 550.00, NULL,
580.00, 'Stockholm Music Festival', '2024-09-05', '2024-09-10', 'Music festival in Stockholm');
```

```

(3, 3, NOW(), '2024-09-10', '2024-09-15', 750.00, 'Pending', 5, '2024-09-08', 700.00, NULL,
725.00, 'Paris Fashion Week', '2024-09-10', '2024-09-15', 'Fashion event in Paris'),
(4, 4, NOW(), '2024-09-12', '2024-09-18', 1500.00, 'Pending', 6, '2024-09-10', 1400.00, NULL,
1450.00, 'Rome Film Festival', '2024-09-12', '2024-09-18', 'Film festival in Rome'),
(5, 5, NOW(), '2024-09-15', '2024-09-20', 450.00, 'Paid', 5, '2024-09-13', 400.00, NULL,
425.00, 'Seoul Food Festival', '2024-09-15', '2024-09-20', 'Food festival in Seoul'),
(6, 6, NOW(), '2024-09-18', '2024-09-22', 560.00, 'Cancelled', 4, '2024-09-16', 500.00,
'2024-09-18', 530.00, 'Tokyo Game Show', '2024-09-18', '2024-09-22', 'Gaming event in Tokyo'),
(7, 7, NOW(), '2024-09-20', '2024-09-25', 650.00, 'Paid', 5, '2024-09-18', 600.00, NULL,
620.00, 'Gothenburg Book Fair', '2024-09-20', '2024-09-25', 'Book fair in Gothenburg'),
(8, 8, NOW(), '2024-09-25', '2024-09-30', 700.00, 'Pending', 5, '2024-09-23', 650.00, NULL,
680.00, 'Munich Oktoberfest', '2024-09-25', '2024-09-30', 'Festival in Munich'),
(9, 9, NOW(), '2024-09-28', '2024-10-03', 800.00, 'Paid', 5, '2024-09-26', 750.00, NULL,
780.00, 'Lyon Lights Festival', '2024-09-28', '2024-10-03', 'Light festival in Lyon'),
(10, 10, NOW(), '2024-10-01', '2024-10-05', 480.00, 'Pending', 4, '2024-09-29', 400.00, NULL,
450.00, 'Milan Fashion Week', '2024-10-01', '2024-10-05', 'Fashion event in Milan'),
(11, 11, NOW(), '2024-10-03', '2024-10-08', 425.00, 'Paid', 5, '2024-10-01', 380.00, NULL,
410.00, 'Cairo Film Festival', '2024-10-03', '2024-10-08', 'Film festival in Cairo'),
(12, 12, NOW(), '2024-10-07', '2024-10-12', 1000.00, 'Pending', 5, '2024-10-05', 950.00,
NULL, 975.00, 'London Literature Festival', '2024-10-07', '2024-10-12', 'Literature festival in
London'),
(13, 13, NOW(), '2024-10-10', '2024-10-15', 1500.00, 'Paid', 5, '2024-10-08', 1400.00, NULL,
1450.00, 'Sydney Opera Festival', '2024-10-10', '2024-10-15', 'Opera festival in Sydney'),
(14, 14, NOW(), '2024-10-12', '2024-10-17', 800.00, 'Pending', 5, '2024-10-10', 750.00, NULL,
780.00, 'New York Film Festival', '2024-10-12', '2024-10-17', 'Film festival in New York'),
(15, 15, NOW(), '2024-10-15', '2024-10-20', 500.00, 'Paid', 5, '2024-10-13', 450.00, NULL,
475.00, 'Toronto Beer Festival', '2024-10-15', '2024-10-20', 'Beer festival in Toronto'),
(16, 16, NOW(), '2024-10-18', '2024-10-22', 1200.00, 'Cancelled', 4, '2024-10-16', 1100.00,
'2024-10-18', 1150.00, 'Zurich International Film Festival', '2024-10-18', '2024-10-22', 'Film
festival in Zurich'),
(17, 17, NOW(), '2024-10-20', '2024-10-25', 650.00, 'Pending', 5, '2024-10-18', 600.00, NULL,
620.00, 'Malmö Food Fair', '2024-10-20', '2024-10-25', 'Food festival in Malmö'),
(18, 18, NOW(), '2024-10-25', '2024-10-30', 350.00, 'Paid', 5, '2024-10-23', 300.00, NULL,
330.00, 'Hamburg Jazz Festival', '2024-10-25', '2024-10-30', 'Jazz festival in Hamburg'),
(19, 19, NOW(), '2024-10-28', '2024-11-02', 1680.00, 'Pending', 5, '2024-10-26', 1600.00,
NULL, 1650.00, 'Florence Art Exhibition', '2024-10-28', '2024-11-02', 'Art exhibition in
Florence'),
(20, 20, NOW(), '2024-11-01', '2024-11-05', 525.00, 'Paid', 4, '2024-10-30', 500.00, NULL,
510.00, 'Nice Film Festival', '2024-11-01', '2024-11-05', 'Film festival in Nice');

```

```

-----
-- Insert test data into the Transaction table
-- Transactions (payments and refunds) linked to bookings
-----

```

```

INSERT INTO Transaction (GuestID, BookingID, Amount, TransactionDate, PaymentMethod,
TransactionType, RefundProcessedDate, Description)
VALUES
(1, 1, 900.00, '2024-09-01', 'CreditCard_Visa', 'Payment', NULL, 'Booking for vacation
rental'),
(2, 2, 600.00, '2024-09-05', 'PayPal', 'Payment', NULL, 'Booking for vacation rental'),
(3, 3, 750.00, '2024-09-10', 'CreditCard_MasterCard', 'Payment', NULL, 'Booking for vacation
rental'),
(4, 4, 1500.00, '2024-09-12', 'BankTransfer', 'Payment', NULL, 'Booking for vacation
rental'),
(5, 5, 450.00, '2024-09-15', 'CreditCard_AMEX', 'Payment', NULL, 'Booking for vacation
rental'),
(6, 6, 560.00, '2024-09-18', 'ApplePay', 'Payment', NULL, 'Booking for vacation rental'),
(7, 7, 650.00, '2024-09-20', 'CreditCard_Visa', 'Payment', NULL, 'Booking for vacation
rental'),
(8, 8, 700.00, '2024-09-25', 'GPay', 'Payment', NULL, 'Booking for vacation rental'),
(9, 9, 800.00, '2024-09-28', 'Maestro', 'Payment', NULL, 'Booking for vacation rental'),
(10, 10, 480.00, '2024-10-01', 'PayPal', 'Payment', NULL, 'Booking for vacation rental'),
(11, 11, 200.00, '2024-09-02', 'CreditCard_Visa', 'Refund', '2024-09-02 10:00:00', 'Partial
refund for early cancellation'),

```

```
(12, 12, 150.00, '2024-09-06', 'PayPal', 'Refund', '2024-09-06 14:00:00', 'Refund due to
service issues'),
(13, 13, 100.00, '2024-09-11', 'CreditCard_MasterCard', 'Refund', '2024-09-11 16:30:00',
'Refund for service issues'),
(14, 14, 1500.00, '2024-09-13', 'BankTransfer', 'Refund', '2024-09-13 12:00:00', 'Full refund
for booking cancellation'),
(15, 15, 450.00, '2024-09-16', 'CreditCard_AMEX', 'Refund', '2024-09-16 10:00:00', 'Full
refund for booking cancellation'),
(16, 16, 560.00, '2024-09-18', 'ApplePay', 'Refund', '2024-09-18 12:00:00', 'Refund for
booking cancellation'),
(17, 17, 650.00, '2024-09-20', 'CreditCard_Visa', 'Refund', '2024-09-20 09:00:00', 'Refund
due to service issues'),
(18, 18, 700.00, '2024-09-25', 'GPay', 'Refund', '2024-09-25 15:00:00', 'Partial refund for
service issues'),
(19, 19, 800.00, '2024-09-28', 'Maestro', 'Refund', '2024-09-28 17:00:00', 'Refund for
booking cancellation'),
(20, 20, 525.00, '2024-10-01', 'PayPal', 'Refund', '2024-10-01 11:30:00', 'Full refund for
booking cancellation');
```

```
-----
-- Insert test data into the TravelAdmin table
-- Admin users responsible for handling reservations and policies
-----
```

```
INSERT INTO TravelAdmin (Name, Email, PhoneNumber, GuestID)
VALUES
('Hans Müller', 'hans.mueller@example.de', '+49 491523456789', 1),
('Emma Karlsson', 'emma.karlsson@example.se', '+46 467021234567', 2),
('Jean Dupont', 'jean.dupont@example.fr', '+33 33122334455', 3),
('Giovanni Rossi', 'giovanni.rossi@example.it', '+39 390123456789', 4),
('Seo-jun Lee', 'seo.jun.lee@example.kr', '+82 821012345678', 5),
('Sakura Tanaka', 'sakura.tanaka@example.jp', '+81 819012345678', 6),
('Anna Svensson', 'anna.svensson@example.se', '+46 46709876543', 7),
('Friedrich Becker', 'friedrich.becker@example.de', '+49 491523456789', 8),
('Pierre Martin', 'pierre.martin@example.fr', '+33 33122334455', 9),
('Alessandro Bianchi', 'alessandro.bianchi@example.it', '+39 390223344556', 10),
('Ahmed El-Sayed', 'ahmed.el-sayed@example.eg', '+20 201234567890', 11),
('John Smith', 'john.smith@example.co.uk', '+44 441234567890', 12),
('Olivia Brown', 'olivia.brown@example.au', '+61 611234567890', 13),
('Michael Johnson', 'michael.johnson@example.ca', '+1 11234567890', 14),
('Emma Williams', 'emma.williams@example.ca', '+1 11234567890', 15),
('Chloe Dubois', 'chloe.dubois@example.ch', '+41 46709876543', 16),
('Mats Andersson', 'mats.andersson@example.se', '+46 46709876544', 17),
('Sophie Schmidt', 'sophie.schmidt@example.de', '+49 491523456789', 18),
('Luca Bruni', 'luca.bruni@example.it', '+39 390234567890', 19),
('Isabelle Laurent', 'isabelle.laurent@example.fr', '+33 33122334455', 20);
```

```
-----
-- Insert test data into the CancellationPolicy table
-- Policies regarding booking cancellations
-----
```

```
INSERT INTO CancellationPolicy (PolicyName, Description)
VALUES
('Flexible', 'Full refund up to 1 day before arrival'),
('Moderate', 'Full refund up to 5 days before arrival'),
('Strict', '50% refund up to 7 days before arrival'),
('Super Strict', '50% refund up to 14 days before arrival'),
('Non-refundable', 'No refund after booking'),
('Relaxed', 'Full refund up to 3 days before arrival'),
('Firm', '75% refund up to 2 days before arrival'),
('Standard', '50% refund up to 4 days before arrival'),
('Special', 'Full refund up to 10 days before arrival'),
('Custom', 'Refund terms are set by the host'),
('Partial Refund', '30% refund up to 3 days before arrival'),
('Full Refund', '100% refund up to 7 days before arrival');
```

```

('Weekend Special', 'No refund on weekends'),
('Holiday Refund', 'Full refund on holidays'),
('Seasonal Refund', 'Partial refund depending on the season'),
('Last Minute', 'No refund for last-minute bookings'),
('High Season', 'Strict refund policy during high season'),
('Event Special', 'No refund during special events'),
('Flexible Plus', 'Full refund up to 2 days before arrival'),
('Summer Special', 'Full refund up to 7 days before arrival');

-----
-- Insert test data into the VacationRentalPolicy table
-- Associating cancellation policies with vacation rentals
-----
INSERT INTO VacationRentalPolicy (VacationRentalID, PolicyID)
VALUES
(1, 1),
(2, 2),
(3, 3),
(4, 4),
(5, 5),
(6, 6),
(7, 7),
(8, 8),
(9, 9),
(10, 10),
(11, 11),
(12, 12),
(13, 13),
(14, 14),
(15, 15),
(16, 16),
(17, 17),
(18, 18),
(19, 19),
(20, 20);

-----
-- Insert test data into the Reservation table
-- Admin-handled reservation details
-----
INSERT INTO Reservation (BookingID, AdminID, DateOfReservation, PaymentStatus, LengthOfStay,
CancellationPolicy, RefundPolicy)
VALUES
(1, 1, '2024-09-01', 'Pending', 7, 'Flexible', 'Full refund up to 1 day before arrival'),
(2, 2, '2024-09-05', 'Paid', 5, 'Moderate', 'Full refund up to 5 days before arrival'),
(3, 3, '2024-09-10', 'Cancelled', 5, 'Strict', '50% refund up to 7 days before arrival'),
(4, 4, '2024-09-12', 'Pending', 6, 'Super Strict', '50% refund up to 14 days before
arrival'),
(5, 5, '2024-09-15', 'Paid', 5, 'Non-refundable', 'No refund after booking'),
(6, 6, '2024-09-18', 'Cancelled', 4, 'Relaxed', 'Full refund up to 3 days before arrival'),
(7, 7, '2024-09-20', 'Paid', 5, 'Firm', '75% refund up to 2 days before arrival'),
(8, 8, '2024-09-25', 'Pending', 5, 'Standard', '50% refund up to 4 days before arrival'),
(9, 9, '2024-09-28', 'Paid', 5, 'Special', 'Full refund up to 10 days before arrival'),
(10, 10, '2024-10-01', 'Pending', 4, 'Custom', 'Refund terms are set by the host'),
(11, 11, '2024-10-03', 'Paid', 5, 'Partial Refund', '30% refund up to 3 days before
arrival'),
(12, 12, '2024-10-07', 'Pending', 5, 'Full Refund', '100% refund up to 7 days before
arrival'),
(13, 13, '2024-10-10', 'Paid', 5, 'Weekend Special', 'No refund on weekends'),
(14, 14, '2024-10-12', 'Pending', 5, 'Holiday Refund', 'Full refund on holidays'),
(15, 15, '2024-10-15', 'Paid', 5, 'Seasonal Refund', 'Partial refund depending on the
season'),
(16, 16, '2024-10-18', 'Cancelled', 4, 'Last Minute', 'No refund for last-minute bookings'),

```

```
(17, 17, '2024-10-20', 'Pending', 5, 'High Season', 'Strict refund policy during high season'),
(18, 18, '2024-10-25', 'Paid', 5, 'Event Special', 'No refund during special events'),
(19, 19, '2024-10-28', 'Pending', 5, 'Flexible Plus', 'Full refund up to 2 days before arrival'),
(20, 20, '2024-11-01', 'Paid', 4, 'Summer Special', 'Full refund up to 7 days before arrival');
```

```
-----
-- Insert test data into the Review table
-- Reviews from guests and hosts
-----
```

```
INSERT INTO Review (BookingID, ReviewerID, ReviewerType, Rating, Comment, ReviewDate)
VALUES
(1, 1, 'Guest', 5, 'Fantastisches Erlebnis, werde definitiv zurückkommen!', '2024-09-01'),
(2, 2, 'Host', 4, 'Bra gäst, men lite bullrigt ibland.', '2024-09-02'),
(3, 3, 'Guest', 5, 'Expérience merveilleuse, très propre!', '2024-09-03'),
(4, 4, 'Host', 3, 'Buon soggiorno, ma il Wi-Fi era un po\' lento.', '2024-09-04'),
(5, 5, 'Guest', 4, '훌륭한 숙박이었어요!', '2024-09-05'),
(6, 6, 'Host', 5, '最高の滞在！また利用したいです。', '2024-09-06'),
(7, 7, 'Guest', 2, 'La habitación era más pequeña de lo esperado.', '2024-09-07'),
(8, 8, 'Host', 4, 'Локация отличная, хорошие гости.', '2024-09-08'),
(9, 9, 'Guest', 5, 'Wonderful stay, highly recommended!', '2024-09-09'),
(10, 10, 'Host', 3, 'Valor razonable, pero con algunos problemas de mantenimiento.', '2024-09-10'),
(11, 11, 'Guest', 5, 'Ottima esperienza!', '2024-09-11'),
(12, 12, 'Host', 4, 'Le client était respectueux et courtois.', '2024-09-12'),
(13, 13, 'Guest', 5, 'Great stay, the place was spotless!', '2024-09-13'),
(14, 14, 'Host', 3, 'L\'invitato è arrivato in ritardo.', '2024-09-14'),
(15, 15, 'Guest', 4, 'Gästvänligt ställe, men något bullrigt.', '2024-09-15'),
(16, 16, 'Host', 5, 'Excelente huésped, muy comunicativo.', '2024-09-16'),
(17, 17, 'Guest', 2, 'El apartamento no cumplió con mis expectativas.', '2024-09-17'),
(18, 18, 'Host', 4, 'Great guest, very tidy and respectful.', '2024-09-18'),
(19, 19, 'Guest', 5, '19-09-2024!', 'حجز رائع، سأعود مرة أخرى'),
(20, 20, 'Host', 3, 'L\'invité aurait pu être plus réactif.', '2024-09-20');
```

```
-----
-- Insert test data into the CustomerService table
-- Customer service issues related to bookings
-----
```

```
INSERT INTO CustomerService (BookingID, IssueDescription, Resolution, ContactMethod, ResolutionDate)
VALUES
(1, 'Late check-in', 'Offered late check-in', 'Phone', '2024-09-02'),
(2, 'WiFi not working', 'Fixed WiFi issue', 'Email', '2024-09-06'),
(3, 'Room not clean', 'Sent cleaning staff', 'Chat', '2024-09-11'),
(4, 'No hot water', 'Repaired hot water system', 'Phone', '2024-09-13'),
(5, 'Key not working', 'Provided new key', 'Email', '2024-09-16'),
(6, 'Noise complaint', 'Offered room change', 'Chat', '2024-09-19'),
(7, 'Room too cold', 'Adjusted thermostat', 'Phone', '2024-09-21'),
(8, 'TV not working', 'Replaced TV', 'Email', '2024-09-26'),
(9, 'No parking space', 'Arranged alternative parking', 'Chat', '2024-09-29'),
(10, 'No towels in room', 'Delivered towels', 'Phone', '2024-10-02'),
(11, 'Noisy neighbors', 'Spoke to neighbors', 'Email', '2024-10-04'),
(12, 'Internet slow', 'Upgraded WiFi', 'Chat', '2024-10-08'),
(13, 'Broken chair', 'Replaced chair', 'Phone', '2024-10-11'),
(14, 'Shower not draining', 'Unclogged drain', 'Email', '2024-10-13'),
(15, 'No hot water', 'Repaired hot water system', 'Chat', '2024-10-17'),
(16, 'Air conditioning not working', 'Fixed AC', 'Phone', '2024-10-20'),
(17, 'No blankets', 'Delivered blankets', 'Email', '2024-10-23'),
(18, 'Broken lock', 'Fixed lock', 'Chat', '2024-10-27'),
(19, 'Water leak', 'Repaired leak', 'Phone', '2024-10-30');
```

```

(20, 'Loud construction noise', 'Offered room change', 'Email', '2024-11-03');

-----
-- Insert test data into the Event table
-- Event details linked to specific bookings
-----
INSERT INTO Event (BookingID, EventName, StartDate, EndDate, Description)
VALUES
(1, 'Berlin Marathon', '2024-09-08', '2024-09-10', 'Annual marathon event'),
(2, 'Stockholm Music Festival', '2024-09-11', '2024-09-13', 'Music festival in Stockholm'),
(3, 'Paris Fashion Week', '2024-09-14', '2024-09-20', 'Fashion event in Paris'),
(4, 'Rome Film Festival', '2024-09-21', '2024-09-25', 'Film festival in Rome'),
(5, 'Seoul Food Festival', '2024-09-26', '2024-09-28', 'Food festival in Seoul'),
(6, 'Tokyo Game Show', '2024-09-29', '2024-10-01', 'Gaming event in Tokyo'),
(7, 'Gothenburg Book Fair', '2024-10-02', '2024-10-05', 'Book fair in Gothenburg'),
(8, 'Munich Oktoberfest', '2024-10-06', '2024-10-10', 'Beer festival in Munich'),
(9, 'Lyon Lights Festival', '2024-10-11', '2024-10-13', 'Light festival in Lyon'),
(10, 'Milan Fashion Week', '2024-10-14', '2024-10-18', 'Fashion event in Milan'),
(11, 'Cairo Film Festival', '2024-10-19', '2024-10-22', 'Film festival in Cairo'),
(12, 'London Literature Festival', '2024-10-23', '2024-10-27', 'Literature festival in
London'),
(13, 'Sydney Opera Festival', '2024-10-28', '2024-11-01', 'Opera festival in Sydney'),
(14, 'New York Comic Con', '2024-11-02', '2024-11-05', 'Comic convention in New York'),
(15, 'Toronto Film Festival', '2024-11-06', '2024-11-10', 'Film festival in Toronto'),
(16, 'Zurich Art Festival', '2024-11-11', '2024-11-14', 'Art festival in Zurich'),
(17, 'Malmo Music Festival', '2024-11-15', '2024-11-17', 'Music festival in Malmo'),
(18, 'Hamburg Christmas Market', '2024-11-18', '2024-11-20', 'Christmas market in Hamburg'),
(19, 'Florence Food Festival', '2024-11-21', '2024-11-24', 'Food festival in Florence'),
(20, 'Nice Jazz Festival', '2024-11-25', '2024-11-27', 'Jazz festival in Nice');

-----
-- Insert test data into the Promotion table
-- Promotional offers for vacation rentals
-----
INSERT INTO Promotion (VacationRentalID, DiscountPercentage, StartDate, EndDate)
VALUES
(1, 10.0, '2024-09-01', '2024-09-07'), -- Hans Müller
(2, 15.0, '2024-09-05', '2024-09-10'), -- Emma Karlsson
(3, 12.5, '2024-09-10', '2024-09-15'), -- Jean Dupont
(4, 20.0, '2024-09-12', '2024-09-18'), -- Giovanni Rossi
(5, 8.0, '2024-09-15', '2024-09-20'), -- Seo-jun Lee
(6, 10.0, '2024-09-18', '2024-09-22'), -- Sakura Tanaka
(7, 7.5, '2024-09-20', '2024-09-25'), -- Anna Svensson
(8, 18.0, '2024-09-25', '2024-09-30'), -- Friedrich Becker
(9, 12.0, '2024-09-28', '2024-10-03'), -- Pierre Martin
(10, 14.0, '2024-10-01', '2024-10-05'), -- Alessandro Bianchi
(11, 9.5, '2024-10-03', '2024-10-08'), -- Ahmed El-Sayed
(12, 11.0, '2024-10-07', '2024-10-12'), -- John Smith
(13, 25.0, '2024-10-10', '2024-10-15'), -- Olivia Brown
(14, 19.0, '2024-10-12', '2024-10-17'), -- Michael Johnson
(15, 15.0, '2024-10-15', '2024-10-20'), -- Emma Williams
(16, 17.5, '2024-10-18', '2024-10-22'), -- Chloe Dubois
(17, 20.0, '2024-10-20', '2024-10-25'), -- Mats Andersson
(18, 5.0, '2024-10-25', '2024-10-30'), -- Sophie Schmidt
(19, 22.0, '2024-10-28', '2024-11-02'), -- Luca Bruni
(20, 10.0, '2024-11-01', '2024-11-05'); -- Isabelle Laurent

-- Enable foreign key checks after all tables are created and data inserted
SET FOREIGN_KEY_CHECKS = 1;

-- Verify data by selecting from the Guest table
SELECT * FROM Guest;

-- Verify data by selecting from the Host table

```

```
SELECT * FROM Host;

-- Verify data by selecting from the VacationRental table
SELECT * FROM VacationRental;

-- Verify data by selecting from the Room table
SELECT * FROM Room;

-- Verify data by selecting from the Amenity table
SELECT * FROM Amenity;

-- Verify data by selecting from the VacationRentalAmenity table
SELECT * FROM VacationRentalAmenity;

-- Verify data by selecting from the CancellationPolicy table
SELECT * FROM CancellationPolicy;

-- Verify data by selecting from the VacationRentalPolicy table
SELECT * FROM VacationRentalPolicy;

-- Verify data by selecting from the Booking table
SELECT * FROM Booking;

-- Verify data by selecting from the Transaction table
SELECT * FROM Transaction;

-- Verify data by selecting from the Review table
SELECT * FROM Review;

-- Verify data by selecting from the CustomerService table
SELECT * FROM CustomerService;

-- Verify data by selecting from the Event table
SELECT * FROM Event;

-- Verify data by selecting from the Promotion table
SELECT * FROM Promotion;

-- Verify data by selecting from the Notification table
SELECT * FROM Notification;

-- Verify data by selecting from the LoginHistory table
SELECT * FROM LoginHistory;

-- Verify data by selecting from the SocialNetwork table
SELECT * FROM SocialNetwork;

-- Verify data by selecting from the GuestSocialNetwork table
SELECT * FROM GuestSocialNetwork;

-- Verify data by selecting from the City table
SELECT * FROM City;

-- Verify data by selecting from the Location table
SELECT * FROM Location;

-- Verify data by selecting from the TravelAdmin table
SELECT * FROM TravelAdmin;

-- Verify data by selecting from the Reservation table
SELECT * FROM Reservation;

-----
-- Data Integrity Check: Orphaned Bookings
```

-- This query checks for bookings that are not linked to any guest.

```
-----
SELECT b.BookingID, b.GuestID, g.Name
FROM Booking b
LEFT JOIN Guest g ON b.GuestID = g.GuestID
WHERE g.GuestID IS NULL;
```

-- Test Case 1: Retrieve guest booking details, host details, and vacation rental details.
 -- Verifies if the join across Booking, Guest, Room, VacationRental, and Host is working as expected.

```
SELECT
    g.Name AS GuestName,
    b.BookingDate,
    b.CheckInDate,
    b.CheckOutDate,
    b.TotalPrice,
    vr.PropertyType,
    h.HostID,
    h.Rating AS HostRating
FROM Booking b
JOIN Guest g ON b.GuestID = g.GuestID
JOIN Room r ON b.RoomID = r.RoomID
JOIN VacationRental vr ON r.VacationRentalID = vr.VacationRentalID
JOIN Host h ON vr.HostID = h.HostID
ORDER BY b.BookingDate DESC;
```

 -- Test Case 2: Retrieve payment and refund details for each booking, along with guest information.

-- Tests the Transaction-Booking-Guest relationship.

```
-----
SELECT
    g.Name AS GuestName,
    b.BookingDate,
    t.Amount AS TransactionAmount,
    t.TransactionDate,
    t.TransactionType,
    t.RefundProcessedDate,
    t.Description AS TransactionDescription
FROM Transaction t
JOIN Booking b ON t.BookingID = b.BookingID
JOIN Guest g ON t.GuestID = g.GuestID
ORDER BY t.TransactionDate DESC;
```

 -- Test Case 3: Retrieve guest reviews, customer service details, and property information.
 -- Tests the Review and CustomerService tables along with Booking, Guest, and VacationRental.

```
-----
SELECT
    g.Name AS GuestName,
    vr.PropertyType,
    r.Rating AS ReviewRating,
    r.Comment AS ReviewComment,
    cs.IssueDescription AS CustomerServiceIssue,
    cs.Resolution AS CustomerServiceResolution,
    cs.ContactMethod AS CustomerServiceContact
FROM Guest g
JOIN Booking b ON g.GuestID = b.GuestID
JOIN Review r ON b.BookingID = r.BookingID
LEFT JOIN CustomerService cs ON b.BookingID = cs.BookingID
JOIN Room rm ON b.RoomID = rm.RoomID
JOIN VacationRental vr ON rm.VacationRentalID = vr.VacationRentalID
JOIN Host h ON vr.HostID = h.HostID
ORDER BY g.Name ASC;
```

```
-----  
-- Test Case 4: Retrieve details for cancelled bookings along with guest information  
-----
```

```
SELECT  
    b.BookingID,  
    g.Name AS GuestName,  
    b.RoomID,  
    b.TotalPrice,  
    b.CancellationRefund,  
    b.DateOfCancellation  
FROM  
    Booking b  
JOIN  
    Guest g ON b.GuestID = g.GuestID  
WHERE  
    b.PaymentStatus = 'Cancelled'  
ORDER BY  
    b.DateOfCancellation DESC;
```

```
-----  
-- Test Case 5: VacationRental Location and City  
-----
```

```
SELECT  
    vr.VacationRentalID,  
    vr.PropertyType,  
    l.PartOfCity,  
    c.CityName,  
    c.Country  
FROM VacationRental vr  
JOIN Location l ON vr.LocationID = l.LocationID  
JOIN City c ON l.CityID = c.CityID  
ORDER BY vr.VacationRentalID;
```

```
-----  
-- Test Case 6: Promotions for Vacation Rentals  
-----
```

```
SELECT  
    vr.PropertyType,  
    p.DiscountPercentage,  
    p.StartDate,  
    p.EndDate  
FROM VacationRental vr  
JOIN Promotion p ON vr.VacationRentalID = p.VacationRentalID  
ORDER BY p.StartDate;
```

```
-- Enable foreign key checks after all tables are created and data inserted  
SET FOREIGN_KEY_CHECKS = 1;
```